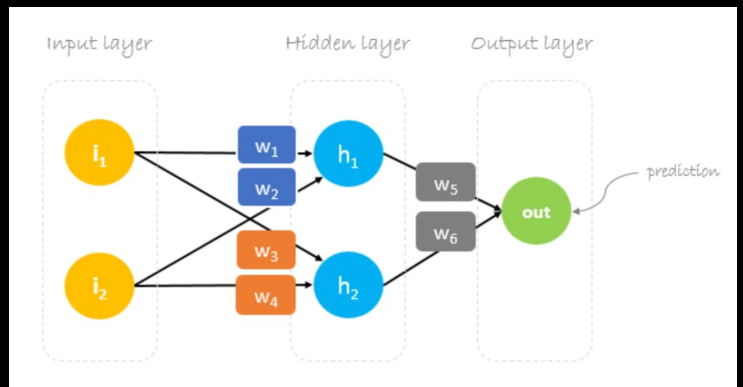


Today's Agenda

1) Forward P

2) Backward P



Inputs

$$x_1 = 0.05$$

$$x_2 = 0.10$$

Weights (I-h)

$$w_1 = 0.15$$

$$w_2 = 0.20$$

$$w_3 = 0.25$$

$$w_4 = 0.30$$

$$w_5 = 0.40$$

$$w_6 = 0.45$$

Bias

$$b_1 = 0.35$$

$$b_2 = 0.60$$

$$\text{Actual } (y) = 0.01 \text{ (y-True)}$$

Forward Propagation

$$z_{h_1} = x_1 w_1 + x_2 w_2 + b_1$$

$$= 0.05 \times 0.15 + 0.1 \times 0.2 + 0.35$$

$$= 0.3775$$

$$z_{h_2} = x_1 w_3 + x_2 w_4 + b_1$$

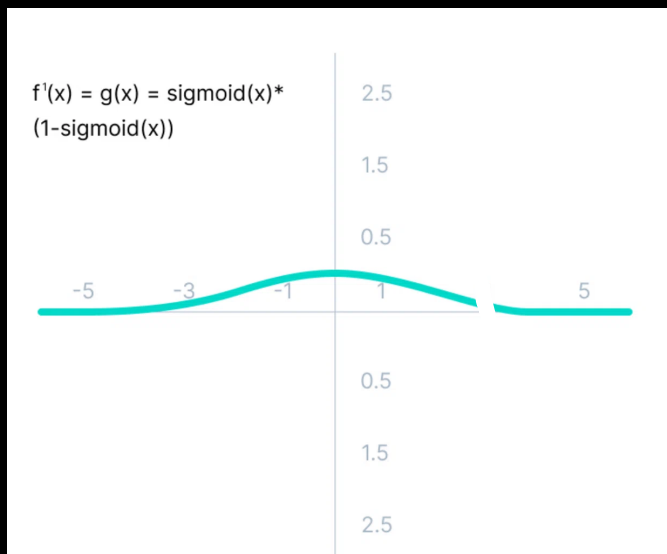
$$= 0.05 \times 0.25 + 0.1 \times 0.3 + 0.35$$

$$= 0.3925$$

Activation

$$a_{h_1} = \text{Sigmoid}(z_{h_1}) = \text{Sigmoid}(0.3775) = 0.5933$$

$$a_{h_2} = \text{Sigmoid}(z_{h_2}) = \text{Sigmoid}(0.3925) = 0.5969$$



$$f(x) = \frac{1}{1 + e^{-x}}$$

Output Layer

$$\begin{aligned} z_{\text{out}} &= a_{h1} \cdot w_5 + a_{h2} \cdot w_6 + b_2 \\ &= 0.5933 \times 0.4 + 0.5969 \times 0.45 + 0.60 \\ &= 1.105 \end{aligned}$$

$$\begin{aligned} a_{z_{\text{out}}} &= \text{Sigmoid}(1.105) \\ &= 0.7514 \end{aligned}$$

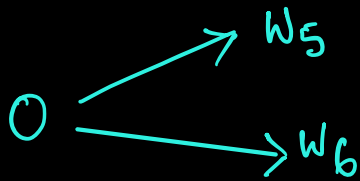
$$\text{Predicted Output } (\hat{y}) = 0.7514$$

Computing the LOSS

MSE

$$\begin{aligned}\text{loss} &= 0.5 (y - \hat{y})^2 \\ &= 0.5 (0.01 - 0.75)^2 \\ &= 0.2748\end{aligned}$$

Backpropagation



w_5 connects with a_{h1} (activated z_{h1})
 w_6 connects with a_{h2} (activated z_{h2})

$\{w_1 - w_6\}$
6 gradients

w_5 (gradient update)

$$\frac{\partial L}{\partial w_5} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial w_5}$$

$$\text{Loss} = 0.5 \left(0.7514 - 0.01 \right)^2$$

$$= 0.27$$

$$\frac{\partial L}{\partial \hat{y}} = (\hat{y} - y)$$

$$= (0.7514 - 0.01) = 0.7414$$

$$\frac{\partial \hat{y}}{\partial z_{\text{out}}} =$$

$$\text{Sigmoid} = \frac{1}{1 + e^{-x}}$$

$$\text{Sigmoid}'(x) = \text{Sigmoid}(x) \cdot (1 - \text{Sigmoid}(x))$$

Constant Rule

$$\frac{d}{dx} (x \cdot c) \xrightarrow{\text{constant}} = c$$

$$z_{\text{out}} = w_5 \times a_{h_1} + w_6 \times a_{h_2} + b$$

$$\left\{ \frac{\partial z_{\text{out}}}{\partial w_5} = a_{h_1} \right\}$$

$$\begin{aligned}
 \frac{\partial \hat{y}}{\partial z_{out}} &= \text{sig} \times (1 - \text{sig}) \\
 &= \hat{y} \times (1 - \hat{y}) \\
 &= 0.7514 \times (1 - 0.7514) \\
 &= 0.1868
 \end{aligned}$$

Final w_5

$$\frac{\partial L}{\partial w_5} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial w_5}$$

$$= 0.7514 \times 0.1868 \times 0.5933$$

$$= 0.0822$$

Weight Update (w_5)

$$\begin{aligned}
 w_{5_{new}} &= w_{old} - \eta \frac{\partial L}{\partial w_5} \\
 &= 0.40 - (0.0001 \times 0.0822) \\
 &= 0.40 - (\quad) \\
 &= 0.399
 \end{aligned}$$

Let's start with w_6

$$\frac{\partial L}{\partial w_6} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial w_6}$$

$$= .7414 \times .1868 \times .5969$$

$$= .0826$$

$$\frac{\partial z_{out}}{\partial w_6} = a_{h_2} = .5969$$

$$\begin{aligned} w_{6_{new}} &= w_{6_{old}} - \eta \frac{\partial L}{\partial w_{6_{old}}} \\ &= 0.45 - (.0001 \times .0826) \\ &= 0.4499 \end{aligned}$$

w_1 gradient

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial a_{h_1}} \cdot \frac{\partial a_{h_1}}{\partial z_{h_1}} \cdot \frac{\partial z_{h_1}}{\partial w_1}$$

$$\frac{\partial z_{out}}{\partial a_{h1}} = w_5 \text{ (constant)}$$

$$z_{out} = w_5 \times a_{h1} + w_6 \times a_{h2} + b$$

$$\frac{\partial z_{out}}{\partial a_{h1}} = 0.40$$

$$\begin{aligned} \frac{\partial a_{h1}}{\partial z_{h1}} &= \text{sig}(a_{h1}) * (1 - \text{sig}(a_{h1})) \\ &= 0.5933 * (1 - 0.5933) \\ &= 0.2413 \end{aligned}$$

$$\begin{aligned} \frac{\partial z_{h1}}{\partial w_1} &= x_1 \quad \left\{ z_{h1} = w_1 \times x_1 + w_2 \times x_2 + b \right. \\ &= 0.05 \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= 0.7414 \times 0.1868 \times 0.40 \times 0.2413 \times 0.05 \\ &= 0.00067 \end{aligned}$$

$$\begin{aligned} w_{1N} &= 0.15 - (0.0001 \times 0.0067) \\ &= 0.1499 \end{aligned}$$

W2 gradient

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial a_{h_1}} \cdot \frac{\partial a_{h_1}}{\partial z \cdot h_1} \cdot \frac{\partial z \cdot h_1}{\partial w_2}$$

$$= 0.7414 \times 0.1868 \times 0.40 \times 0.2413 \times 0.10$$

$$\frac{\partial L}{\partial w_2} = 0.00135 \quad \{w_5, w_6, w_1, w_2\}$$

$$w_{2N} = 0.20 - (0.0001 \times 0.00135) \underline{\underline{w_4, w_3}}$$

gradient update (w4)

$$\frac{\partial L}{\partial w_4} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial a_{h_2}} \cdot \frac{\partial a_{h_2}}{\partial z \cdot h_2} \cdot \frac{\partial z \cdot h_2}{\partial w_4}$$

$$\frac{\partial z_{out}}{\partial z \cdot h_2} = w_6 = 0.45$$

$$\frac{\partial a_{h_2}}{\partial z \cdot h_2} = a_{h_2} * (1 - a_{h_2})$$

$$= 0.5969 * (1 - 0.5969)$$

$$\frac{\partial a_{h2}}{\partial z_{h2}} = 0.2406$$

$$\frac{\partial z_{h2}}{\partial w_4} = x_2 = 0.10$$

$$\frac{\partial L}{\partial w_4} = 0.7414 \times 0.1868 \times 0.45 \times 0.2406 \times 0.10$$

$$= 0.001499$$

$$w_{4N} = 0.30 - (0.0001 \times 0.001499) = 0.29$$

W3 Gradient Update

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial a_{h2}} \cdot \frac{\partial a_{h2}}{\partial z_{h2}} \cdot \frac{\partial z_{h2}}{\partial w_3}$$

$$\frac{\partial z_{h2}}{\partial w_3} = x_1 = 0.05$$

$$\frac{\partial L}{\partial w_3} = 0.7414 \times 0.1868 \times 0.45 \times 0.2406 \times 0.05$$

$$= 0.0007497$$

$$w_{3N} = 0.25 - (0.0001 \times 0.0007497)$$

NEW

$$w_1 = .1499$$

$$w_2 = .1999$$

$$w_3 = .2499$$

$$w_4 = 0.299$$

$$w_5 = 0.399$$

$$w_6 = 0.4499$$

$$= .2499$$

Bias Update

From Output (b_2)

$$z_{out} = w_5 \cdot a_{h1} + w_6 \cdot a_{h2} + b_2$$

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial \tilde{y}} \cdot \frac{\partial \tilde{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial b_2}$$

$$= .7414 \times .1868 \times \frac{1}{1} = 0.1385$$

$$\frac{\partial z_{out}}{\partial b_2} = 1$$

Hidden layer Bias (b_1)

$$z_{h_1} = +b_1$$

$$z_{h_2} = +b_1$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial b_1 \text{ from } h_1} + \frac{\partial L}{\partial b_1 \text{ from } h_2}$$

Gradient bias from h_1

$$\frac{\partial L}{\partial b_1 h_1} = \frac{\partial L}{\partial \tilde{y}} \cdot \frac{\partial \tilde{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial a_{h_1}} \cdot \frac{\partial a_{h_1}}{\partial z_{h_1}} \cdot \frac{\partial z_{h_1}}{\partial b_1 h_1}$$

$$= 0.7414 \times 0.1868 \times 0.4 \times 0.2413 \times 1$$

$$= 0.013367$$

$$\frac{\partial z_{out}}{\partial a_{h_1}} = w_5 = 0.40$$

$$\frac{\partial a_{h_1}}{\partial z_{h_1}} = 0.5933 \times (1 - 0.5933)$$

$$= 0.2413$$

$$\frac{\partial z_{h1}}{\partial b_{1h1}} = \underline{1}$$

Bias (h_2)

$$\frac{\partial L}{\partial b_{1h2}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_{out}} \cdot \frac{\partial z_{out}}{\partial a_{h2}} \cdot \frac{\partial a_{h2}}{\partial z_{h2}} \cdot \frac{\partial z_{h2}}{\partial b_{1h2}}$$

$$= .7414 \times .1868 \times .45 \times .2406 \times 1$$

$$= .01499$$

$$\frac{\partial z_{out}}{\partial a_{h2}} = w_6 = 0.45$$

$$\frac{\partial a_{h2}}{\partial z_{h2}} = .5969 \times (1 - .5969)$$

$$= .2406$$

$$\frac{\partial z_{h2}}{\partial b_{1h2}} = 1$$

Final b_1

$$b_1 = .01499 + .01336$$

$$= .0283$$

Bias Update

$$B_N = B_0 - \eta \frac{\partial L}{\partial B_0}$$

$$b_{2N} = 0.60 - (0.0001 \times 0.1385)$$

$$= 0.59986$$

$$b_{1N} = 0.35 - (0.0001 \times 0.0283)$$

$$= 0.3499$$